MILITARY SPECIFICATION SHEET

MICROPHONES, LINEAR M49199/4

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-49199

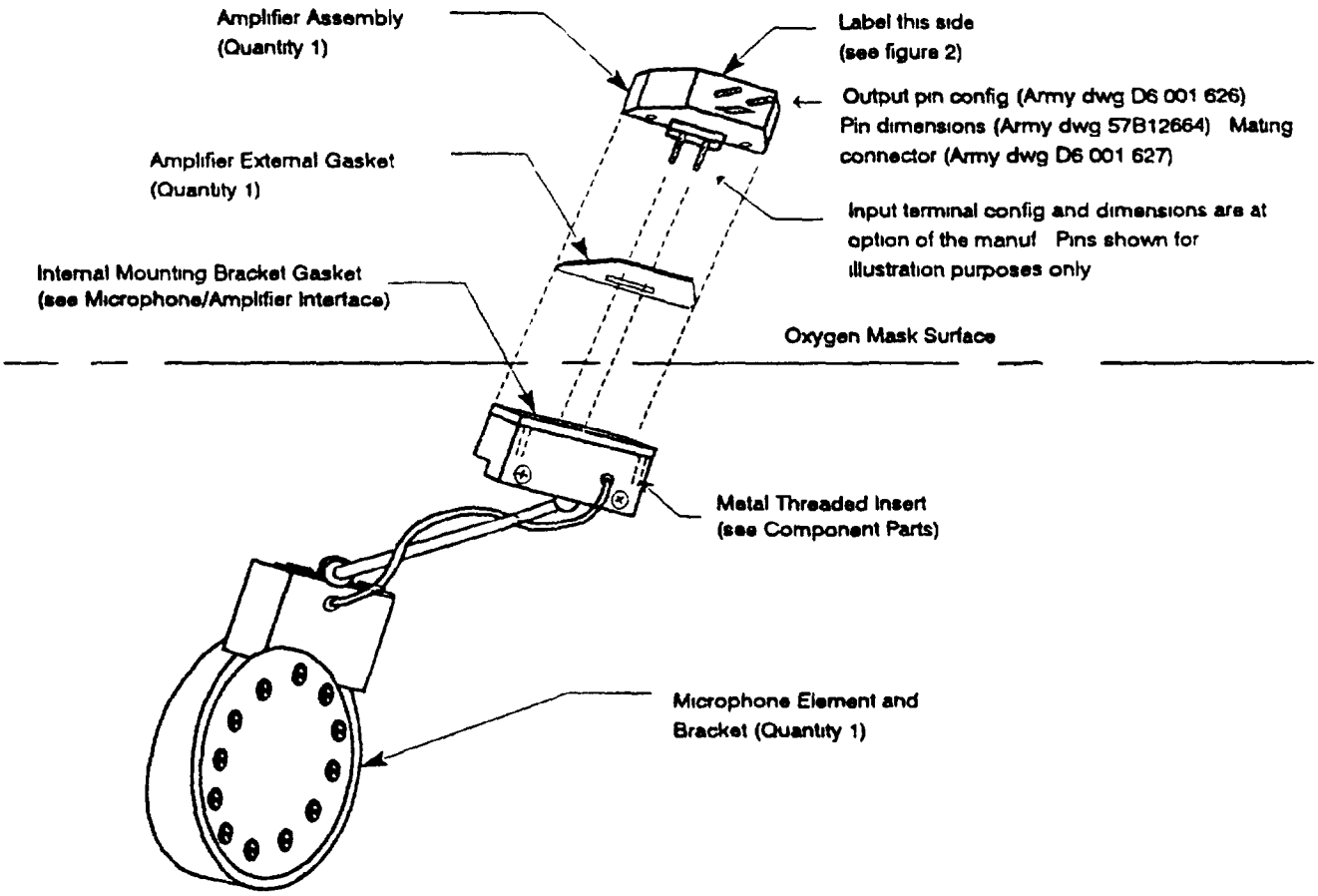


FIGURE 1. Microphone assembly, P/N M49199/4-01 (exploded view)

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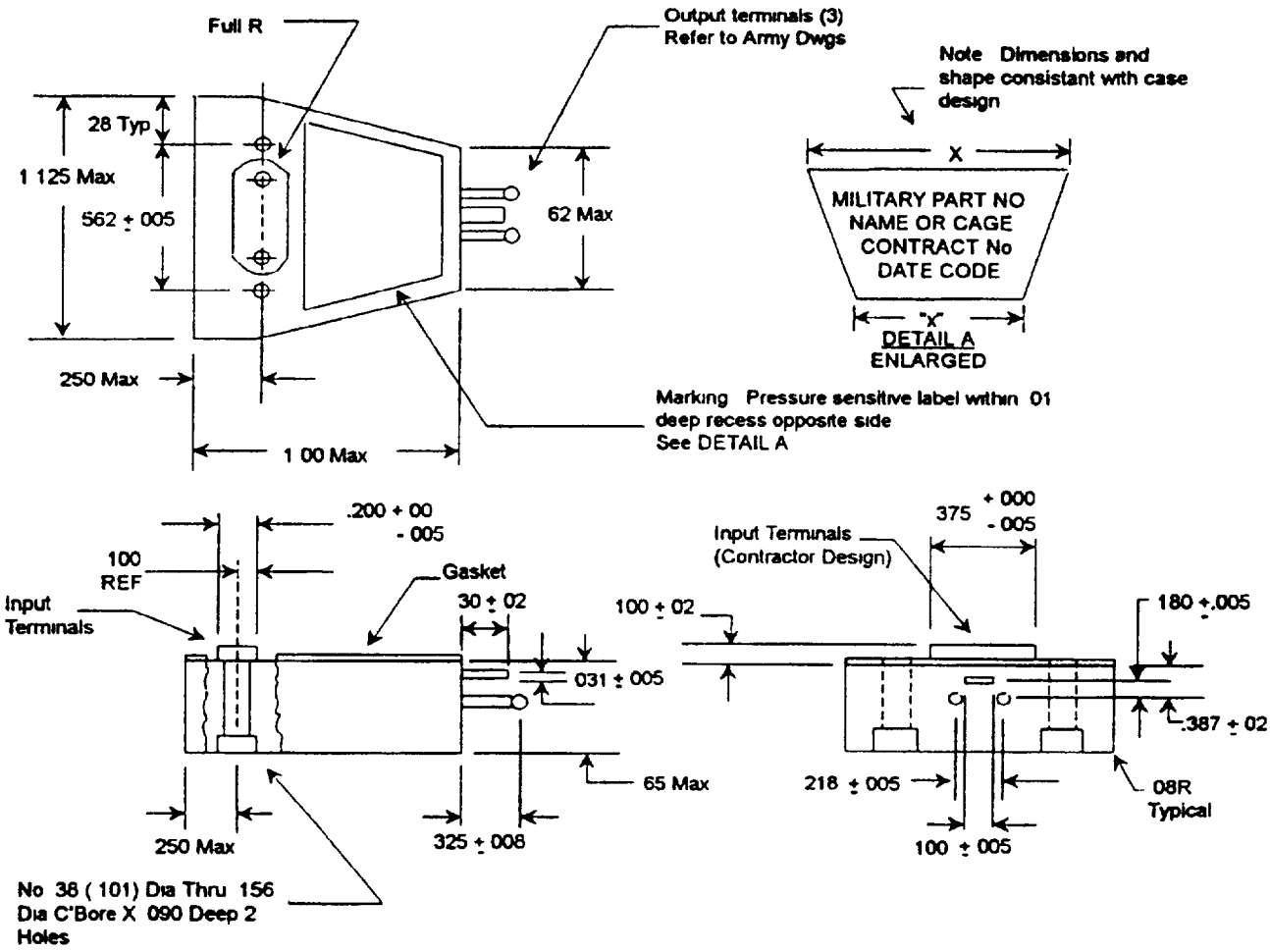


FIGURE 2 Amplifier assembly

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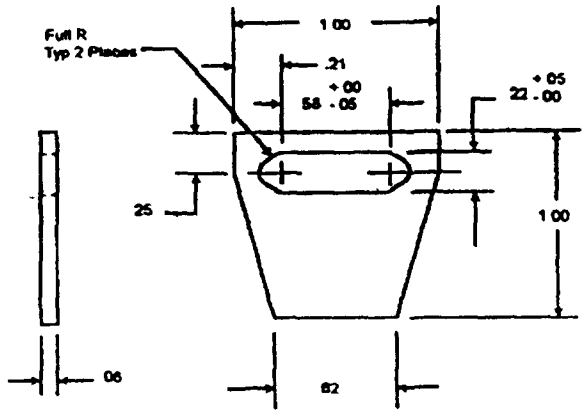


FIGURE 3a Amplifier external gasket

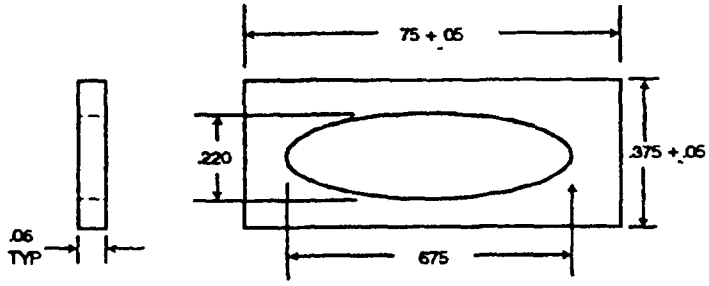


FIGURE 3b Amplifier internal gasket

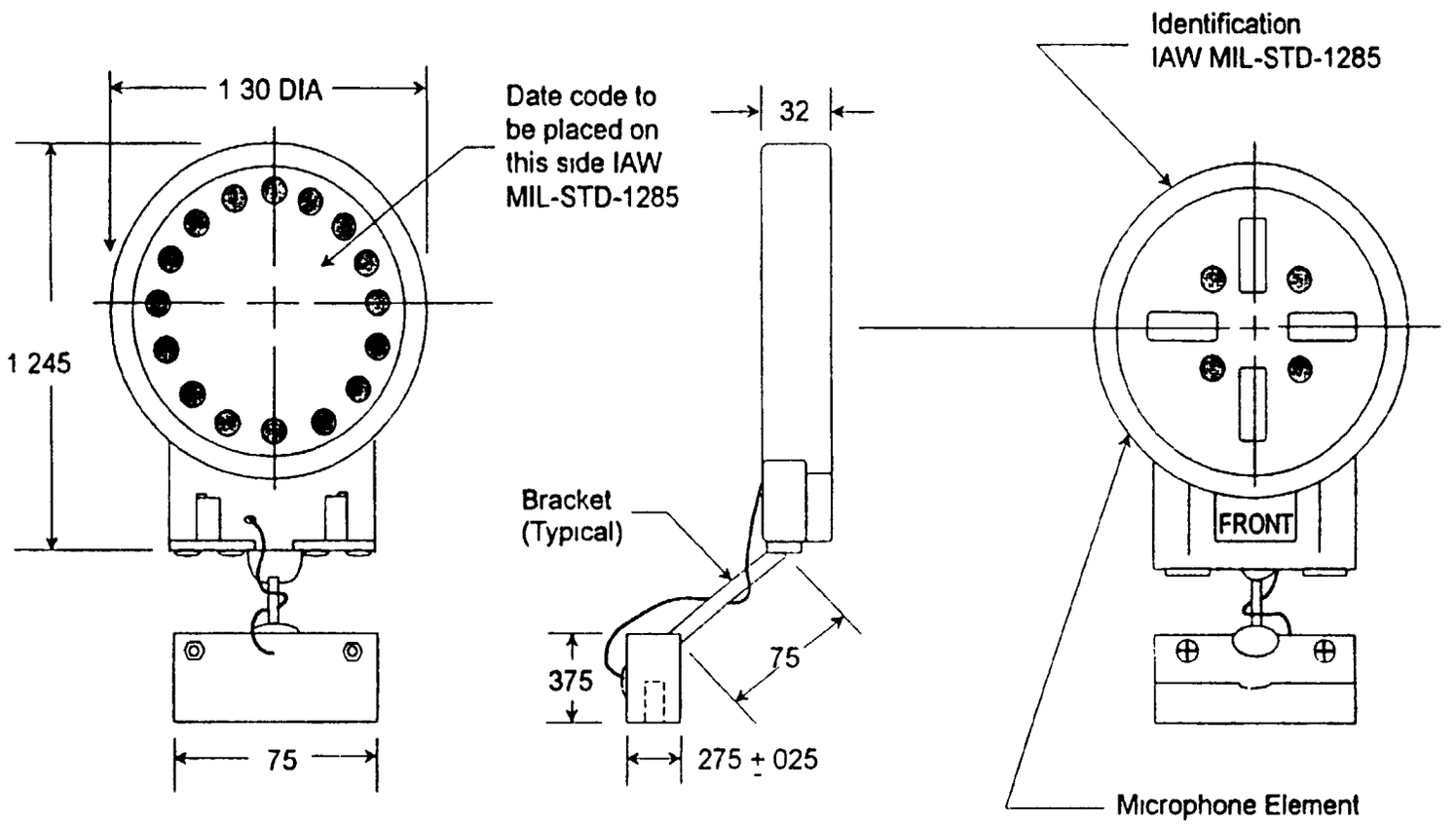
NOTES

1 All dimensions in Figure 3a are maximum except where indicated. Actual gasket dimensions shall be consistent with amplifier case design

2 Material: Silicone Rubber, commercial grade

FIGURE 3 Amplifier gaskets

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NOTE Microphone configuration and dimensions are typical, and may vary Size shoule be minimized to the maximum extent feasible

FIGURE 4 Microphone element and bracket

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REQUIREMENTS:

DESIGN and CONSTRUCTION: See Figures 1 through 4.

MICROPHONE/AMPLIFIER INTERFACE The microphone amplifier interface at the oxygen mask surface shall seal to protect the electrical contacts from severe moisture effects inside the oxygen mask. The internal gasket (Figure 3b) shall be permanently attached to the mounting bracket.

OUTPUT PINS· The output terminals shall contain two pins made in accordance with Air Force Drawing 57B12664 and a third pin in accordance with Army drawing D6 001 626 The pins shall be arranged as detailed in Army drawing D6 001 626 Output terminals shall interface with the jack detailed in Army drawing D6 001 627

INPUT PINS Input terminal configuration is not specified since pins are within system design and not required to be compatible with external connectors. The connector design shall be within envelope dimensions shown for compatibility with oxygen mask design

CONFIGURATION The microphone assembly is comprised of a preamplifier assembly and a microphone element assembly The microphone element assembly shall contain the microphone element, and only the electronics necessary for transfer of the audio signal to the preamplifier. To the maximum extent practical, all electronics shall be contained within the preamplifier assembly rather than the microphone element assembly to minimize the risk of a fire potential within the oxygen mask Voltages and currents in the microphone element assembly shall not create sparks nor any other adverse effect which presents a fire potential inside the oxygen mask

COMPONENT PARTS The microphone shall be supplied with the following

One gasket (see figure 3a) which shall be attached to the bottom of the amplifier via adhesive The gasket shall be made of silicone rubber, commercial grade

Two MS35275-210 machine screws, fillister head (2-56 x 3/4 lg) The amplifier shall not be damaged by application of 8 in-lb torque to the screws

WEIGHT 62g max for the assembly

COLOR· Semigloss, lusterless, black

SOLDERABILITY. Soldering shall be in accordance with MIL-STD-2000

VOLTAGE OPERATING RANGE 12 - 30 volts DC, 27 5 VDC nominal

INPUT CURRENT 17 mA max at 27.5 VDC when connected as shown in Figure 5

IMPEDANCE· The electrical impedance of the microphone assembly shall be such that the unit shall be capable of meeting the sensitivity and frequency response to a close sound source while connected to a 100 ohm ±1% load impedance.

SENSITIVITY. 96 4 minimum at 104 dB SPL with 27 5 VDC supply voltage, and shall not vary by more than 1 0 dB when the supply voltage is varied between 22 to 30 volts

SENSITIVITY AT REDUCED BIAS VOLTAGE Shall not vary by more than 12 dB at 104 dB SPL from the 27.5 VDC established value when the supply voltage is varied between 12 and 22 volts

SENSITIVITY AT ALTITUDE: ±3 dB of sensitivity at ground level while at a simulated altitude of 15,000 ft above sea level

FREQUENCY RESPONSE AT GROUND LEVEL See Figure 6

FREQUENCY RESPONSE AT ALTITUDE Within ±5 dB of ground level measurement

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NOISE IMMUNITY. See Figure 7.

LINEARITY Resistive load shall be 100 ±1% ohms

GAIN STABILIZATION·

Within 100 mS when supply voltage is between 23 and 30 VDC

Within 150 mS when supply voltage is between 12 and 22 VDC

ENVIROMENTAL STRESS SCREENING (ESS) 24 hours of temperature cycling between -54℃ and 55℃, with each cycle consisting of 1 hour at each temperature and the change between temperature occuring at a minimum rate of 5℃ per minute

OVERVOLTAGE PROTECTION 15 minutes operation in test circuit shown in Figure 5 while the 330 ohm resistor is shorted

MARKING

M49199/4-01
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER

EMI Electromagnetic environmental test levels shall be as defined in the following tables

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TABLE I MIL-STD-461 RS03 electromagnetic environmental test levels

FREQUENCY	SIGNAL	MODULATION	FIELD 3/
14 KHz - 1 MHz	CW	NONE	20 V/M avg
1 MHz - 18 GHz	CW	NONE	200 V/M avg
18 GHz - 40 GHz	CW	NONE	20 V/M avg
1/ 1 MHz - 18 GHz	PAM	Table 1	200 V/M peak
18 GHz - 40 GHz	PAM	Table 1	20 V/M peak

TABLE II. Worst case modulation over required frequency band.

STEP	FREQUENCY SUB-BAND	INITIAL SET FREQUENCY	FIELD (PEAK)	MODULATION (SQUARE WAVE)
2/ A	1.0 MHz - 10.0 MHz	5.0 MHz	200 V/M	30 Hz, 300 Hz, 3 KHz, 30 KHz
B	10.0 MHz - 100.0 MHz	50.0 MHz	200 V/M	30 Hz, 300 Hz, 3 KHz, 30 KHz
C	100.0 MHz - 1.0 GHz	500.0 MHz	200 V/M	30 Hz, 300 Hz, 3 KHz, 30 KHz
D	1 0 GHz - 10 0 GHz	SEE D_1 AND D_2 WHERE APPLICABLE		
D_1	1.0 GHz - 8.0 GHz	5.0 GHz	200 V/M	3 KHz, 30 KHz, 300 KHz
D_2	8.0 GHz - 10 0 GHz	9.0 GHz	200 V/M	3 KHz, 30 KHz, 300 KHz
E	10 0 GHz - 18 0 GHz	15.0 GHz	200 V/M	3 KHz, 30 KHz, 300 KHz
(4) F	18.0 GHz - 40 GHz	30.0 GHz	20 V/M	300 Hz, 3 KHz, 30 KHz

NOTES:

1/ Tests from 1.0 MHz - 18.0 GHz and 18.0 GHz - 40.0 GHz shall be performed by sub-bands defined in Table II for worst case modulation conditions If no susceptibility exists in the sub-band specified, the modulation in Table III shall be used. Note 2/ below, defines the procedures for determining modulation required

2/ EXAMPLE: Step A: (1.0 - 10 0 MHz)
(a) Adjust to initial set frequency of 5.0 MHz
(b) Perform RS03 Tests @ 200 V/M for each modulation specified in Table II
(c) Using worst case modulation determined in STEP (b), perform tests over frequency sub-bands @ 200 V/M. If no susceptibility condition was encountered in STEP (b), use modulation specified in Table III for 1.0 - 10 0 MHz

TABLE III. Modulation fall back values.

SQUARE WAVE (50% DUTY CYCLE)
300 Hz
300 Hz
250 Hz
See D_1 or D_2
100 KHz
250 KHz
60 KHz
2 KHz

3/ NOTE

$$\frac{V}{M}\ (P)\ \sqrt{DutyCycle} = \frac{V}{M}(AV)$$

For CW

$$V/M\ (P)\ rms = V/M\ (AV)\ rms$$

4/ Except for 33-34 GHz where the field peak intensity is 200 V/M

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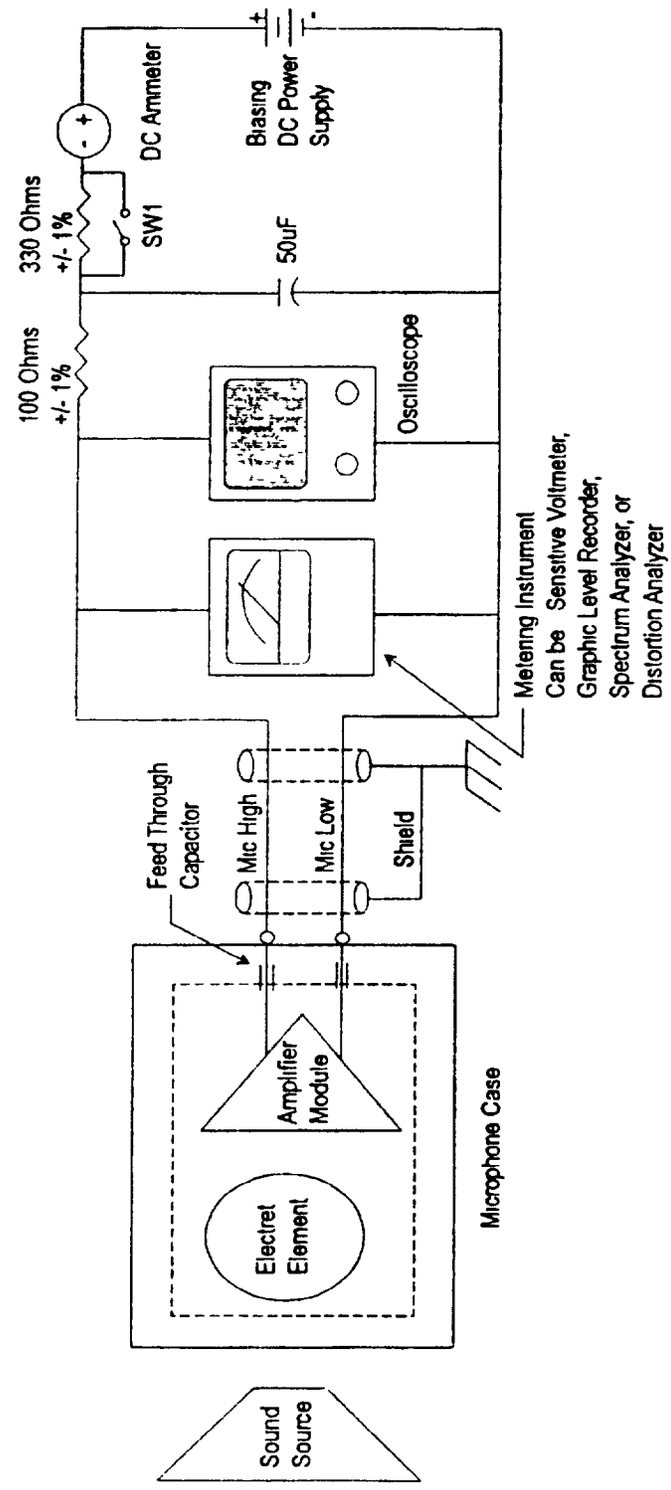


FIGURE 5 Test circuit schematic

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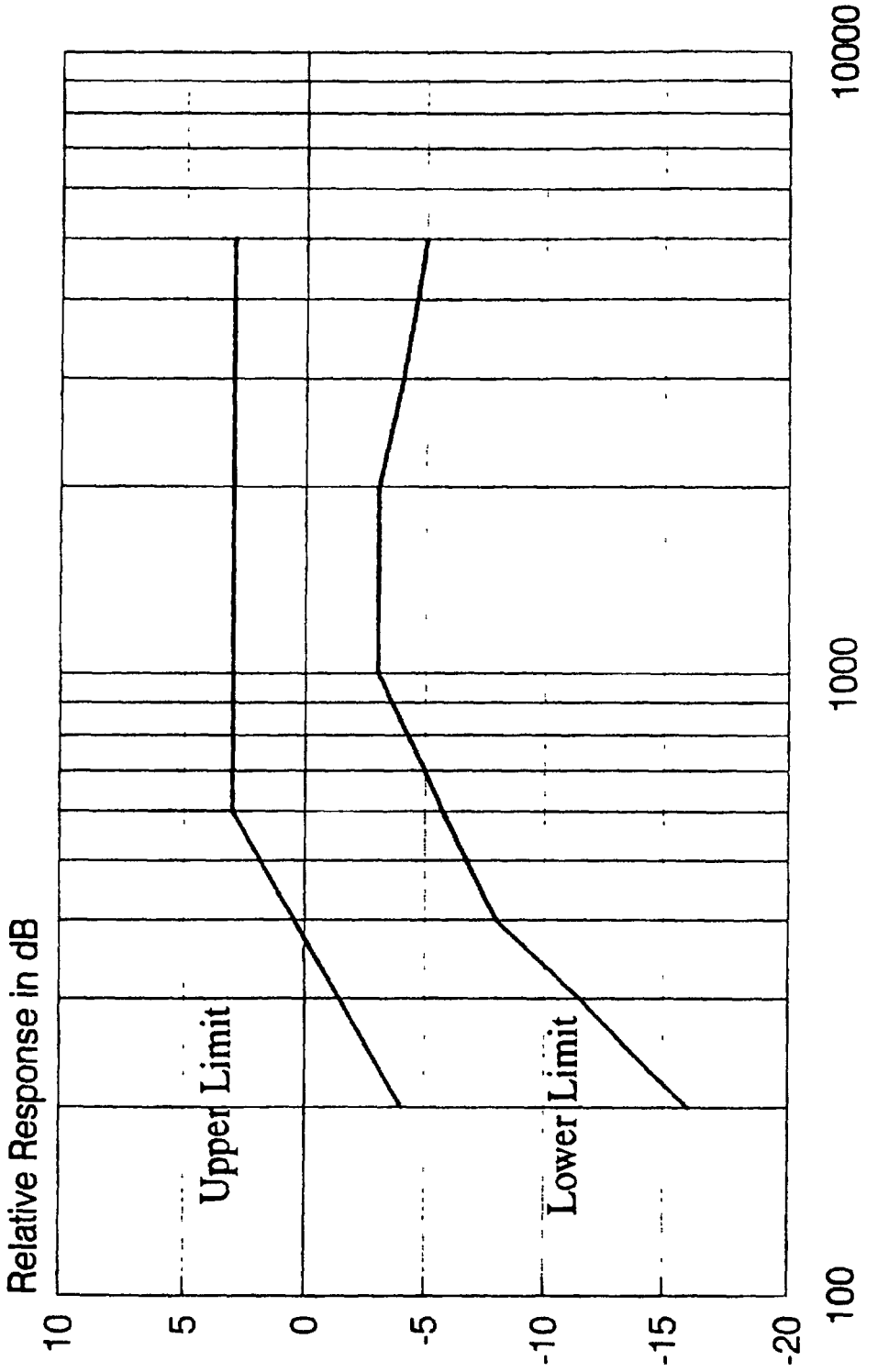


FIGURE 6. Frequency response - limit curves

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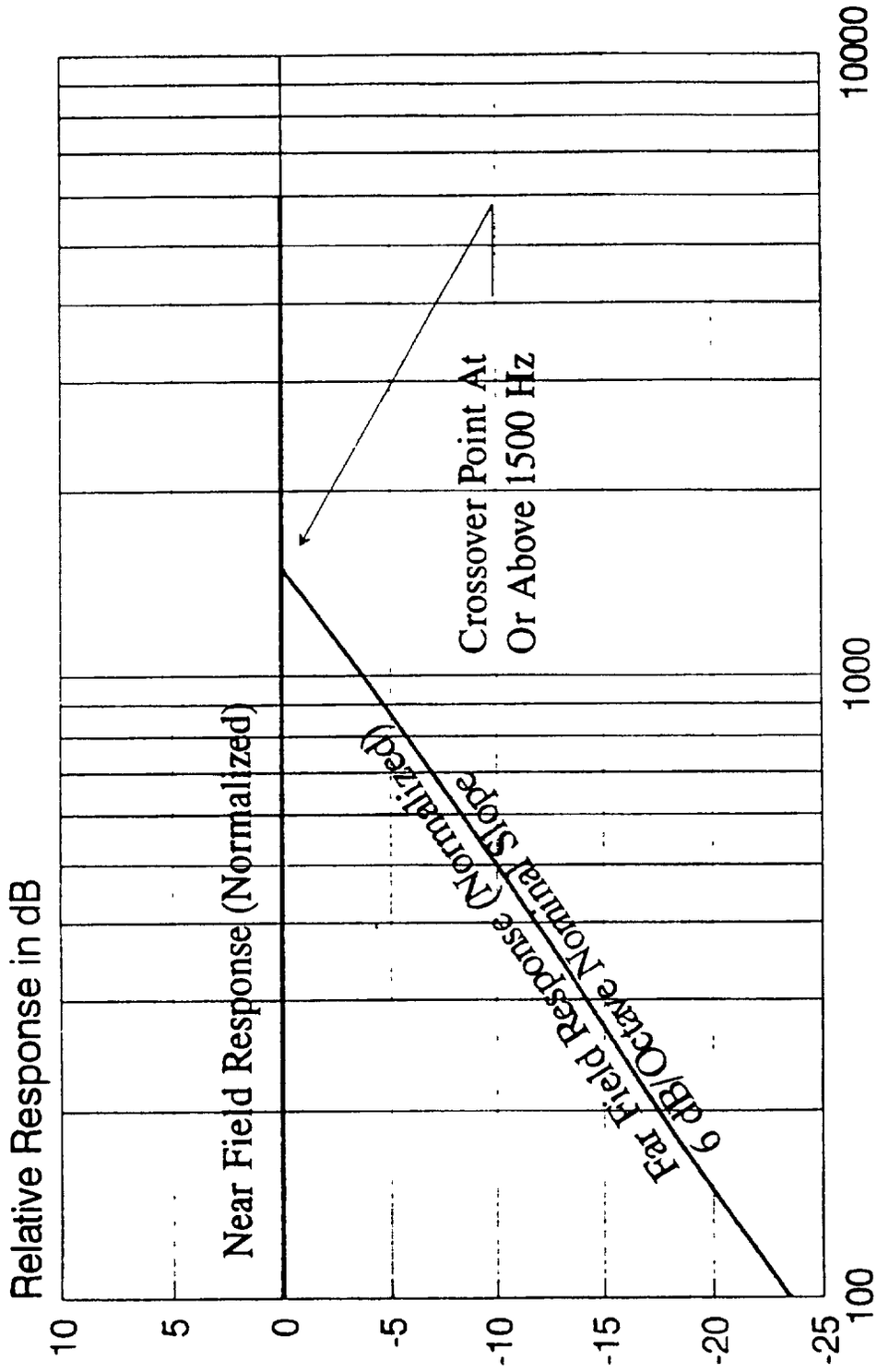


FIGURE 7 Noise-immunity characteristics

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INTENDED USE. Microphone, identified as linear, M49199/4-01, is a noise-canceling, DC-powered, high-gain microphone designed for use in a pressure-type oxygen mask The microphone is intended for use as a part of the airborne communication system which will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft The M49199/4-01 microphone is designed to work with intercommunication system AIC-14, specifically control box C-2645D The preservation, packing, and marking specified herein are intended for direct shipments to the Government However, this specification may also be used for the preparation of microphones for shipment from the parts contractor to the original equipment manufacturer

Preparing Activity
Navy - AS

(Project 5965-0220-04)

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